Argan Fire Watch

Security Assessment & Remediation Report

Penetration Test Findings, Vulnerability Analysis
& Full Backend Hardening Documentation

Project	Argan Fire Watch — Wildfire Monitoring System
Date	6 April 2026
Scope	Backend API Security
Status	**All Critical Issues Remediated**
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Security Team Internal Document

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Executive Summary

This report documents the complete security lifecycle for the **Argan Fire Watch** backend: from live penetration test findings to full code-level remediation. The system was subjected to a comprehensive black-box penetration test, during which **10 critical/high vulnerabilities** were identified and subsequently fixed. Additionally, three backend hardening tasks (*log integrity*, *RBAC*, and *SQL injection protection*) were applied as part of a parallel security audit.

Vulnerability Count by Severity

Severity	Discovered	Status
CRITICAL	7	FIXED
HIGH	3	FIXED
MEDIUM	3	FIXED

Key Fixes Applied

- **Authentication Enforcement** — JWT tokens required on all sensitive endpoints
- **Authorization / RBAC** — Role-based decorators and ownership checks
- **Input Validation** — Strict whitelist validation on sensor data submissions
- **Rate Limiting** — Brute-force protection on all authentication endpoints
- **Log Integrity** — Hash-chain + HMAC signatures with append-only enforcement
- **SQL Injection Protection** — Replacement of f-strings with parameterized queries
- **Security Headers** — Added X-Frame-Options, X-Content-Type-Options, etc.
- **CORS Hardening** — Configured allowlist instead of dangerous wildcard
- **JWT Secret Enforcement** — Mandatory environment variable, no defaults

Penetration Test Findings

This section presents the vulnerabilities discovered during the live penetration test, with their attack type, impact analysis, and proof-of-concept evidence.

Finding #1 — Unauthenticated Admin API Access CRITICAL

Vulnerability Details

Category:	Broken Authentication / Missing Access Control
CVSS Score:	9.8 (Critical)
OWASP:	A01:2021 — Broken Access Control
Endpoint:	`GET /api/admin/cooperatives/pending`
Tool used:	`curl`

Description: The entire admin blueprint was publicly accessible without any token or role verification. An unauthenticated attacker could retrieve the full list of pending cooperatives, including sensitive PII (emails, phone numbers, real names), approve or reject cooperative accounts, and access all administrative statistics.

Data Exposed:

- Cooperative details: internal ID, address, region, creation date
- Owner PII: full name, email, phone number
- User list: ID, full name, email, role (admin/firefighter/cooperative owner)
- Global statistics: active alerts, active sensors, pending approvals

⚠ Proof of Concept — Unauthorized Request Revealing Sensitive Data

```
┌──(root㉿kali)-[~]
└─# curl -v "http://172.20.10.2:5000/api/admin/cooperatives/pending"
*   Trying 172.20.10.2:5000...
* Established connection to 172.20.10.2 (172.20.10.2 port 5000) from 10.0.2.15 port 60560
* using HTTP/1.x
> GET /api/admin/cooperatives/pending HTTP/1.1
> Host: 172.20.10.2:5000
> User-Agent: curl/8.17.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Server: Werkzeug/3.1.8 Python/3.11.0
< Date: Sun, 05 Apr 2026 10:17:31 GMT
< Content-Type: application/json
< Content-Length: 292
< Connection: close
<
[{"_id":4,"address":"Temsia","createdAt":"2026-03-10","name":"Abdelmalek's coop","ownerEmail":"abdelmalekhamda@gmai.com","ownerFirstName":"WRWEF","ownerLastName":"FEEWFE","ownerName":"WRWEF FEEWFE","region":"Inezgane-Ait-Melloul","statu
t":"pending","telephone":"0622222222","zoneName":null}]
* shutting down connection #0

┌──(root㉿kali)-[~]
└─#
```

Finding #2 — Unauthenticated Sensor Data Injection CRITICAL

Vulnerability Details

Category:	Broken Authentication + Data Tampering
CVSS Score:	9.8 (Critical)
OWASP:	A01:2021 + A03:2021
Endpoint:	`POST /api/mesures`
Tool used:	`curl`

Description: The sensor measurement endpoint accepted POST requests with no authentication and no input validation. Any attacker could inject arbitrary sensor values, including extreme temperatures (e.g. `temperature=999`), which immediately appeared in charts and could trigger false fire alerts via WhatsApp notifications.

⚠ Proof of Concept — Injecting fake 999°C reading

```
┌──(root㉿kali)-[~]
└─# curl -X POST "http://192.168.56.1:5000/api/mesures" \
-H "Content-Type: application/json" \
-d '{"id_capteur":1,"temperature":999,"humidite":0,"vent":999}'
{"message":"Mesure ajout\u00e9e"}

┌──(root㉿kali)-[~]
```

Finding #3 — Sensor Fleet Information Disclosure CRITICAL

Vulnerability Details

Category:	Unauthenticated Information Disclosure
CVSS Score:	9.8 (Critical)
OWASP:	A01:2021
Endpoint:	`GET /api/sensors`

Description: The sensor listing endpoint exposed GPS coordinates, real-time environmental readings, signal quality, device models, cooperative names, and zone names to any unauthenticated user. This allows an attacker to map the entire sensor network and identify coverage gaps for targeted arson.

⚠ Proof of Concept — Direct Unauthorized Access to Admin Endpoints

```
└─# curl -s "$T/api/sensors" | head -c 400
[{"altitude":null,"connectivity":"excellent","cooperative":"Abde's coop","coo
":"-9.598100","model":null,"signalQuality":100,"status":"ACTIF","temperature"
```

Finding #4 — Firefighter Dashboard Exposure CRITICAL

Vulnerability Details

Category:	Sensitive Operational Data Exposure
CVSS Score:	9.8 (Critical)
OWASP:	A01:2021
Endpoint:	`GET /api/dashboard/pompier/stats`

Description: Real-time firefighter operational data was publicly accessible, including the number of active alerts, ongoing fires, available firefighters, and monitored zones. This enables an attacker to time real attacks during low-staffing periods or saturate the system with false alerts.

⚠ Proof of Concept — Unauthorized Access to Firefighter Operational Statistics

```
┌──(root㉿kali)-[~]
└─# curl -s "$T/api/dashboard/pompier/stats"
{"alertesActives":3,"alertesResolues":16,"alertesTotal":19,"incendiesEnCours"
:0,"pompiersDisponibles":3,"zonesSurveillees":0}

┌──(root㉿kali)-[~]
└─#
```

Finding #5 — Cooperative Chart IDOR HIGH

Vulnerability Details

Category:	Insecure Direct Object Reference (IDOR)
CVSS Score:	High
OWASP:	A01:2021
Endpoint:	`GET /api/cooperative/<coop_id>/charts/*`

Description: Chart endpoints had no authentication and no ownership verification. Any attacker could access temperature, humidity, and alert charts for any cooperative by guessing sequential IDs (1, 2, 3...), enabling corporate espionage and infrastructure reconnaissance.

⚠ Proof of Concept — Unauthorized Access and Disclosure of Cooperative Data

```
┌──(root㉿kali)-[~]
└─# curl -s "$T/api/cooperative/1/charts/temperature"
[{"avgTemp":39.4,"humidity":44.2,"maxTemp":55.4,"name":"Wed"},{"avgTemp":999.
0,"humidity":0.0,"maxTemp":999.0,"name":"Sun"}]

┌──(root㉿kali)-[~]
└─#
```

Finding #6 — Missing Security Headers MEDIUM

Vulnerability Details

Category:	Security Misconfiguration
CVSS Score:	Medium
OWASP:	A05:2021
Tool used:	Nikto v2.5.0

Nikto scan revealed:

- `X-Frame-Options` header absent → clickjacking risk
- `X-Content-Type-Options` header absent → MIME sniffing / XSS risk
- `Server` header exposes `Werkzeug/3.1.8 Python/3.11.0` → version leak

Proof of Concept — Missing Security Headers Leading to Browser-Side Vulnerabilities

```
┌──(root㉿kali)-[~]
└─# ikto - headers de securite
nikto -h http://192.168.56.1:5000 -Format txt -o nikto_out.txt
Command 'ikto' not found, did you mean:
  command 'nikto' from deb nikto
Try: apt install <deb name>
- Nikto v2.5.0
---------------------------------------------------------------------------
+ Target IP:          192.168.56.1
+ Target Hostname:    192.168.56.1
+ Target Port:        5000
+ Start Time:         2026-04-05 12:48:22 (GMT1)
---------------------------------------------------------------------------
+ Server: Werkzeug/3.1.8 Python/3.11.0
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/e
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the conte
ing-content-type-header/
+ No CGI Directories found (use '-C all' to force check all possible dirs)
+ 8101 requests: 0 error(s) and 2 item(s) reported on remote host
+ End Time:           2026-04-05 12:49:27 (GMT1) (65 seconds)
---------------------------------------------------------------------------
+ 1 host(s) tested
```

...

Finding #7 — CORS Wildcard Origin MEDIUM

Vulnerability Details

Category:	Security Misconfiguration
CVSS Score:	Medium
OWASP:	A05:2021

Description: CORS was configured with `origins: "*"` combined with `supports_credentials: True`. This combination violates the W3C CORS specification and enables private network scanning from any external website.

Finding #8 — No Rate Limiting on Auth Endpoints HIGH

Vulnerability Details

Category:	Identification and Authentication Failures
CVSS Score:	High
OWASP:	A07:2021

Description: Email verification codes (6 digits = 1,000,000 combinations) had no rate limiting. At 100 requests/second (trivial with threading), an attacker could brute-force any verification code in approximately 2.8 hours.

Proof of Concept — Brute-Force of Verification Codes Due to Missing Rate Limiting

```python
# ==========================
# SEND VERIFICATION CODE
# ==========================
Windsurf: Refactor | Explain | Generate Docstring | ×
@auth_bp.route("/send-verification", methods=["POST"])
def send_verification():
    # Rate limiting check
    ip = request.remote_addr
    allowed, _ = current_app.check_rate_limit(ip)
    if not allowed:
        return {"message": "Too many requests. Please try again later."}, 429

    data = request.get_json()
    if not data:
        return {"message": "Invalid data"}, 400

    email = data.get("email")
    if not email:
        return {"message": "Email is required"}, 400

    # Check if email already exists in database
    conn = get_db_connection()
    cursor = conn.cursor(dictionary=True)
    try:
        cursor.execute("SELECT id_utilisateur FROM utilisateurs WHERE email=%s", (email,))
        if cursor.fetchone():
            return {"message": "Email already registered"}, 409
    finally:
        cursor.close()
        conn.close()
```

Finding #9 — Hidden Unauthenticated Notifications Endpoint HIGH

Vulnerability Details

Category:	Hidden Endpoint Enumeration / Information Disclosure
CVSS Score:	High
OWASP:	A01:2021
Tool used:	`ffuf` with `dirb/common.txt`

Description: `ffuf` discovered `/api/notifications` returning HTTP 200 with real-time alert data to any unauthenticated user, revealing ongoing incident details and operational patterns.

⚠ Proof of Concept — Hidden Endpoint Enumeration Revealing Real-Time Incident Data

```
┌──(root㉿kali)-[~]
└─# ffuf - decouvrir routes cachees
ffuf -u "$T/api/FUZZ" \
-w /usr/share/wordlists/dirb/common.txt \
-mc 200,401,403
Encountered error(s): 2 errors occurred.
        * -u flag or -request flag is required
        * Either -w or --input-cmd flag is required

Fuzz Faster U Fool - v2.1.0-dev
```

```
Session  Actions  Edit  View  Help
EXAMPLE USAGE:
  Fuzz file paths from wordlist.txt, match all responses but filter out those with content-size 42.
  Colored, verbose output.
    ffuf -w wordlist.txt -u https://example.org/FUZZ -mc all -fs 42 -c -v

  Fuzz Host-header, match HTTP 200 responses.
    ffuf -w hosts.txt -u https://example.org/ -H "Host: FUZZ" -mc 200

  Fuzz POST JSON data. Match all responses not containing text "error".
    ffuf -w entries.txt -u https://example.org/ -X POST -H "Content-Type: application/json" \
      -d '{"name": "FUZZ", "anotherkey": "anothervalue"}' -fr "error"

  Fuzz multiple locations. Match only responses reflecting the value of "VAL" keyword. Colored.
    ffuf -w params.txt:PARAM -w values.txt:VAL -u https://example.org/?PARAM=VAL -mr "VAL" -c

  More information and examples: https://github.com/ffuf/ffuf

Encountered error(s): 2 errors occurred.
        * -u flag or -request flag is required
        * Either -w or --input-cmd flag is required


        /'___\  /'___\           /'___\
       /\ \__/ /\ \__/  __  __  /\ \__/
       \ \ ,__\\ \ ,__\/\ \/\ \ \ \ ,__\
        \ \ \_/ \ \ \_/\ \ \_\ \ \ \ \_/
         \ \_\   \ \_\  \ \____/  \ \_\
          \/_/    \/_/   \/___/    \/_/

       v2.1.0-dev
________________________________________________

 :: Method           : GET
 :: URL              : http://172.20.10.2:5000/api/FUZZ
 :: Wordlist         : FUZZ: /usr/share/wordlists/dirb/common.txt
 :: Follow redirects : false
 :: Calibration      : false
 :: Timeout          : 10
 :: Threads          : 40
 :: Matcher          : Response status: 200,401,403
________________________________________________

notifications           [Status: 200, Size: 444, Words: 13, Lines: 2, Duration: 312ms]
:: Progress: [4614/4614] :: Job [1/1] :: 199 req/sec :: Duration: [0:00:22] :: Errors: 0 ::

┌──(root㉿kali)-[~]
```

Finding #10 — Hardcoded JWT Secret Default CRITICAL

Vulnerability Details

Category:	Cryptographic Failures / Hardcoded Credentials
CVSS Score:	Critical
OWASP:	A02:2021

Description: The JWT secret fell back to a hardcoded string `"jwt-default-secret"` if the environment variable was not set. An attacker knowing this public default can forge valid JWT tokens for any user ID with any role, achieving complete authentication bypass.

Proof of Concept — Forging JWT Tokens via Hardcoded Default Secret

```
JWT_SECRET = os.getenv("JWT_SECRET_KEY", "jwt-default-secret")
JWT_EXPIRATION_HOURS = int(os.getenv("JWT_EXPIRATION_HOURS", "24"))
```

Security Audit Findings (Code Review)

In parallel with the penetration test, a source-code audit identified three additional backend hardening issues.

Audit Finding A — Log Integrity CRITICAL

Problem: The `logs_securite` table had no tamper-evidence mechanism. Logs were recalculated on-the-fly without stored hash chains or HMAC signatures, and no `BEFORE UPDATE`/`BEFORE DELETE` triggers blocked modification or deletion.

Impact: An attacker (or malicious insider) could silently erase or modify evidence of fire events and unauthorized access.

Proof of Concept — Tampering and Deletion of Security Logs Without Integrity Controls

```sql
-- Table structure for table `logs_securite`
--

CREATE TABLE `logs_securite` (
  `id_log` bigint UNSIGNED NOT NULL,
  `action` varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL,
  `table_cible` varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL,
  `id_enregistrement` varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL,
  `ancienne_valeur` json DEFAULT NULL,
  `nouvelle_valeur` json DEFAULT NULL,
  `adresse_ip` varchar(45) COLLATE utf8mb4_unicode_ci DEFAULT NULL,
  `horodatage` timestamp(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3),
  `id_utilisateur` int UNSIGNED DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

```
DELIMITER $$
CREATE TRIGGER `trg_audit_utilisateurs` AFTER UPDATE ON `utilisateurs` FOR EACH ROW BEGIN

    INSERT INTO logs_securite (
        action,
        table_cible,
        id_enregistrement,
        ancienne_valeur,
        nouvelle_valeur
    )
    VALUES (
        'UPDATE',
        'utilisateurs',
        CAST(OLD.id_utilisateur AS CHAR),
        JSON_OBJECT(
            'email',  OLD.email,
            'statut', OLD.statut,
            'nom',    OLD.nom,
            'prenom', OLD.prenom
        ),
        JSON_OBJECT(
            'email',  NEW.email,
            'statut', NEW.statut,
            'nom',    NEW.nom,
            'prenom', NEW.prenom
        )
    );

END
$$
DELIMITER ;
```

Audit Finding B — Insufficient RBAC CRITICAL

Problem: Admin routes (`/users`, `/stats`, `/logs`, `/alerts`) had no `@token_required` or `@admin_required` decorators. Cooperative endpoints used `coop_id` from the URL without verifying that the requesting user was the actual owner (`id_responsable`).

Proof of Concept — Bypassing Role-Based Access Control on Admin and Cooperative Endpoint

```
# GET - toutes les demandes pending
Windsurf: Refactor | Explain | Generate Docstring | X
@admin_bp.route("/cooperatives/pending", methods=["GET"])
def get_pending():
    conn = get_db_connection()
    cursor = conn.cursor(dictionary=True)
```

```sql
DELIMITER $$
CREATE TRIGGER `trg_audit_utilisateurs` AFTER UPDATE ON `utilisateurs` FOR EACH ROW BEGIN

    INSERT INTO logs_securite (
        action,
        table_cible,
        id_enregistrement,
        ancienne_valeur,
        nouvelle_valeur
    )
    VALUES (
        'UPDATE',
        'utilisateurs',
        CAST(OLD.id_utilisateur AS CHAR),
        JSON_OBJECT(
            'email',  OLD.email,
            'statut', OLD.statut,
            'nom',    OLD.nom,
            'prenom', OLD.prenom
        ),
        JSON_OBJECT(
            'email',  NEW.email,
            'statut', NEW.statut,
            'nom',    NEW.nom,
            'prenom', NEW.prenom
        )
    );

END
$$
DELIMITER ;
```

Audit Finding C — SQL Injection via f-strings MEDIUM

Problem: Multiple routes in `admin.py`, `cooperative_routes.py`, and `zones.py` built SQL queries using Python f-strings with dynamic column names and interval expressions instead of parameterized queries.

Proof of Concept — SQL Injection Exploitation via Python f-Strings

```python
cursor.execute(f"""
    SELECT
        DATE(date_creation) AS date,
        {group_by} AS period,
        COUNT(*) AS count
    FROM utilisateurs
    WHERE date_creation >= DATE_SUB(NOW(), {interval})
    GROUP BY {group_by}, DATE(date_creation)
    ORDER BY date_creation ASC
""")
```

```python
cursor.execute(f"""
    SELECT
        DATE(date_creation) AS date,
        {group_by} AS period,
        SUM(CASE WHEN statut = 'approved' THEN 1 ELSE 0 END) AS approved,
        SUM(CASE WHEN statut = 'pending' THEN 1 ELSE 0 END) AS pending,
        SUM(CASE WHEN statut = 'rejected' THEN 1 ELSE 0 END) AS rejected
    FROM cooperatives
    WHERE date_creation >= DATE_SUB(NOW(), {interval})
    GROUP BY {group_by}, DATE(date_creation)
    ORDER BY date_creation ASC
""")
```

```
cursor.execute(f"""
    SELECT
        DATE(date_installation) AS date,
        {group_by} AS period,
        COUNT(*) AS count
    FROM capteurs
    WHERE date_installation >= DATE_SUB(NOW(), {interval})
    GROUP BY {group_by}, DATE(date_installation)
    ORDER BY date_installation ASC
""")
```

Remediations

Fix #1 — Admin API Authentication FIXED

File: `backend/app/routes/admin.py`

The `@admin_required` decorator was added to every route in the admin blueprint, enforcing both JWT validation and ADMIN role check before any data is returned.

Before — Vulnerable

```
@admin_bp.route("/cooperatives/pending", methods=["GET"])
def get_pending():
    # NO AUTHENTICATION CHECK
    conn = get_db_connection()
    cursor = conn.cursor(dictionary=True)
    cursor.execute("""
        SELECT c.id_cooperative, c.nom_cooperative, c.region,
               u.nom, u.prenom, u.email, u.telephone
        FROM cooperatives c
        JOIN utilisateurs u ON c.id_responsable = u.id_utilisateur
        WHERE c.statut = 'pending'
    """)
    return jsonify(cursor.fetchall()), 200  # PII returned to
   anyone
```

After — Fixed

```
from app.utils.auth import admin_required

@admin_bp.route("/cooperatives/pending", methods=["GET"])
@admin_required  # <- Token + ADMIN role enforced
def get_pending():
    conn = get_db_connection()
    cursor = conn.cursor(dictionary=True)
    cursor.execute("""
        SELECT c.id_cooperative, c.nom_cooperative, c.region,
               u.nom, u.prenom, u.email, u.telephone
        FROM cooperatives c
        JOIN utilisateurs u ON c.id_responsable = u.id_utilisateur
        WHERE c.statut = 'pending'
    """)
    return jsonify(cursor.fetchall()), 200
```

Applied to all 5 admin routes: `/cooperatives/pending`, `/cooperatives/<id>/approve`, `/cooperatives/<id>/reject`, `/cooperatives`, `/cooperatives/<id>`.

Fix #2 — Sensor Data Injection + Input Validation FIXED

File: `backend/app/routes/mesures_routes.py`

Added `@token_required` plus strict whitelist validation on all four input fields.

Before — Vulnerable

```python
@mesures_bp.route("/mesures", methods=["POST"])
def ajouter_mesure():
    # NO AUTHENTICATION, NO VALIDATION
    data = request.json
    conn = get_db_connection()
    cursor = conn.cursor()
    cursor.callproc("sp_ajouter_mesure", (
        data["id_capteur"],  # Could be: -1, "abc", null
        data["temperature"], # Could be: 999, -999, "hot"
        data["humidite"],    # Could be: 150, -50, "wet"
        data["vent"]         # Could be: 500, -100, "fast"
    ))
    conn.commit()
    return jsonify({"message": "Mesure ajoutee"}), 200  # Always accepts
```

After — Fixed

```python
from app.utils.auth import token_required

@mesures_bp.route("/mesures", methods=["POST"])
@token_required  # <- JWT required
def ajouter_mesure():
    data = request.json
    if not data:
        return jsonify({"error": "Invalid request body"}), 400

    # Validate id_capteur: must be positive integer
    try:
        id_capteur = int(data.get("id_capteur"))
        if id_capteur <= 0:
            return jsonify({"error": "Invalid id_capteur"}), 400
    except (ValueError, TypeError):
        return jsonify({"error": "id_capteur must be an integer"}), 400

    # Validate temperature: realistic forest range -50 to 85 deg C
    try:
        temperature = float(data.get("temperature"))
        if temperature < -50 or temperature > 85:
            return jsonify({"error": "temperature must be -50 to 85"}), 400
    except (ValueError, TypeError):
        return jsonify({"error": "temperature must be a number"}), 400

    # Validate humidity: 0-100%
    try:
        humidite = float(data.get("humidite"))
        if humidite < 0 or humidite > 100:
            return jsonify({"error": "humidite must be 0-100"}), 400
    except (ValueError, TypeError):
        return jsonify({"error": "humidite must be a number"}), 400
```

```python
    # Validate wind: 0-200 km/h
    try:
        vent = float(data.get("vent"))
        if vent < 0 or vent > 200:
            return jsonify({"error": "vent must be 0-200"}), 400
    except (ValueError, TypeError):
        return jsonify({"error": "vent must be a number"}), 400

    conn = get_db_connection()
    cursor = conn.cursor()
    try:
        cursor.callproc("sp_ajouter_mesure",
                        (id_capteur, temperature, humidite, vent))
        conn.commit()
        return jsonify({"message": "Mesure ajoutee"}), 201
    except Exception:
        conn.rollback()
        return jsonify({"error": "Internal error"}), 500
```

Fix #3 — Sensor Fleet Information Disclosure FIXED

File: `backend/app/routes/sensors.py`

Added `@token_required` plus role-based data filtering at the database level.

Before — Vulnerable

```python
@sensors_bp.route("/sensors", methods=["GET"])
def get_sensors():  # NO AUTHENTICATION
    conn = get_db_connection()
    cursor = conn.cursor(dictionary=True)
    cursor.execute("""
        SELECT
            c.latitude, c.longitude,  /* GPS exposed
            c.altitude_m,
            m.temperature_c,          /* Real-time readings exposed
            m.humidite_pct,
            m.vitesse_vent_kmh,
            m.qualite_signal,     /* Infrastructure status exposed
            co.nom_cooperative,   /* Organization structure exposed
            z.nom_zone
        FROM capteurs c
        LEFT JOIN mesures m ON ...
        LEFT JOIN zones_forestieres z ON ...
        LEFT JOIN cooperatives co ON ...
    """)
    return jsonify(cursor.fetchall())  # All GPS data to anyone
```

After — Fixed

```python
from app.utils.auth import token_required

@sensors_bp.route("/sensors", methods=["GET"])
@token_required  # <- JWT required
def get_sensors():
    user = getattr(request, "current_user", None)
```

```python
    conn = get_db_connection()
    cursor = conn.cursor(dictionary=True)

    if user and user.get("role") in ["COOPERATIVE","UTILISATEUR_COOP","COOP"]:
        # Cooperative users: only see their own sensors
        cursor.execute(
            "SELECT id_cooperative FROM cooperatives "
            "WHERE id_responsable = %s", (user.get("user_id"),))
        coop_row = cursor.fetchone()
        if not coop_row:
            return jsonify([]), 200
        cursor.execute("""
            SELECT DISTINCT c.id_capteur, c.latitude, c.longitude,
                   m.temperature_c, m.qualite_signal,
                   co.nom_cooperative, z.nom_zone
            FROM capteurs c
            LEFT JOIN mesures m ON m.id_capteur = c.id_capteur
            WHERE z.id_cooperative = %s
            ORDER BY co.nom_cooperative
        """, (coop_row["id_cooperative"],))
    else:
        # ADMIN: see all sensors (authorized)
        cursor.execute("""
            SELECT DISTINCT c.id_capteur, c.latitude, c.longitude,
                   m.temperature_c, m.qualite_signal,
                   co.nom_cooperative, z.nom_zone
            FROM capteurs c
            LEFT JOIN mesures m ON m.id_capteur = c.id_capteur
            ORDER BY co.nom_cooperative
        """)
    return jsonify(cursor.fetchall()), 200
```

Fix #4 — Firefighter Dashboard FIXED

File: `backend/app/routes/pompier.py`

Added `@pompier_required` (accepts roles: `POMPIER`, `CHEF_EQUIPE`, `ADMIN`) to all three dashboard routes and the notifications endpoint.

Before — Vulnerable

```python
@pompier_bp.route("/dashboard/pompier/stats", methods=["GET","OPTIONS"])
def get_dashboard_stats():  # NO AUTHENTICATION
    ...
    return jsonify({"alertesActives": alertes_actives,
                    "pompiersDisponibles": pompiers_dispo, ...}), 200
```

After — Fixed

```python
from app.utils.auth import pompier_required

@pompier_bp.route("/dashboard/pompier/stats", methods=["GET","
```

```
    OPTIONS"])
@pompier_required  # <- POMPIER / CHEF_EQUIPE / ADMIN only
def get_dashboard_stats():
    ...
    return jsonify({...}), 200
```

Fix #5 — Cooperative Chart IDOR FIXED

File: `backend/app/routes/cooperative_routes.py`

Added `@cooperative_required` plus an explicit ownership check against the database.

Before — Vulnerable

```
@coop_bp.route("/cooperative/<int:coop_id>/charts/temperature",
               methods=["GET"])
def get_temperature_chart(coop_id):  # NO AUTH, NO OWNERSHIP CHECK
    cursor.execute(
        "SELECT id_capteur FROM capteurs c "
        "JOIN zones_forestieres z ON c.id_zone = z.id_zone "
        "WHERE z.id_cooperative = %s", (coop_id,))
    ...
    return jsonify(chart_data)  # Returns data to ANYONE
```

After — Fixed

```
from app.utils.auth import cooperative_required

@coop_bp.route("/cooperative/<int:coop_id>/charts/temperature",
               methods=["GET"])
@cooperative_required  # <- Auth check
def get_temperature_chart(coop_id):
    user = getattr(request, "current_user", None)
    if user and user.get("role") not in ["ADMIN"]:
        cursor.execute(
            "SELECT id_responsable FROM cooperatives "
            "WHERE id_cooperative = %s", (coop_id,))
        coop = cursor.fetchone()
        # Ownership check: requesting user must own this
    cooperative
        if not coop or coop["id_responsable"] != user["user_id"]:
            return jsonify({"error": "Unauthorized"}), 403
    ...
```

Applied to: `/charts/temperature`, `/charts/alerts`, `/charts/sensors`, and `/dashboard`.

Fix #6 — Security Headers FIXED

File: `backend/app/__init__.py`

Before — Vulnerable

```
def create_app():
    app = Flask(__name__, ...)
```

```
    CORS(app, ...)
    app.secret_key = "arganfirewatch"
    # NO after_request hook for security headers
    # NO Server header removal
    return app
```

After — Fixed

```
def create_app():
    app = Flask(__name__, ...)
    CORS(app, ...)
    app.secret_key = os.getenv("FLASK_SECRET_KEY", "arganfirewatch-default")

    @app.after_request
    def add_security_headers(response):
        response.headers["X-Frame-Options"] = "DENY"
        response.headers["X-Content-Type-Options"] = "nosniff"
        response.headers["Referrer-Policy"] = "strict-origin-when-cross-origin"
        response.headers["X-XSS-Protection"] = "1; mode=block"
        if "Server" in response.headers:
            del response.headers["Server"]  # Hide Werkzeug version
        return response

    return app
```

Header	Value	Protection
`X-Frame-Options`	`DENY`	Prevents clickjacking
`X-Content-Type-Options`	`nosniff`	Prevents MIME sniffing/XSS
`Referrer-Policy`	`strict-origin-when-cross-origin`	Privacy protection
`X-XSS-Protection`	`1; mode=block`	Legacy browser XSS filter
`Server`	*(removed)*	Hides framework version

Fix #7 — CORS Wildcard Origin FIXED

File: `backend/app/__init__.py`

Before — Vulnerable

```
# INVALID per W3C spec: wildcard + credentials is forbidden
CORS(app,
     resources={r"/api/*": {"origins": "*"}},  # WILDCARD - dangerous
     supports_credentials=True)
```

After — Fixed

```
# Read allowed origins from environment variable
allowed_origins = os.getenv(
```

```
    "ALLOWED_ORIGINS", "http://localhost:5173").split(",")

CORS(app,
     resources={r"/api/*": {"origins": allowed_origins}},  # Explicit list
     supports_credentials=True,  # Now valid (not wildcard)
     allow_headers=["Content-Type", "Authorization"],
     methods=["GET", "POST", "PUT", "PATCH", "DELETE", "OPTIONS"])

# .env configuration:
# ALLOWED_ORIGINS=http://localhost:5173,https://app.example.com
```

Fix #8 — Rate Limiting on Auth Endpoints FIXED

File: `backend/app/routes/auth_routes.py` & `backend/app/__init__.py`

After — Rate Limiter Implementation

```
from collections import defaultdict
import time

RATE_LIMIT_STORE = defaultdict(list)
RATE_LIMIT_REQUESTS = 10    # Max 10 requests
RATE_LIMIT_WINDOW   = 300   # Per 5 minutes

def check_rate_limit(ip_address):
    now = time.time()
    cutoff = now - RATE_LIMIT_WINDOW
    # Remove timestamps outside the window
    RATE_LIMIT_STORE[ip_address] = [
        ts for ts in RATE_LIMIT_STORE[ip_address] if ts > cutoff
    ]
    if len(RATE_LIMIT_STORE[ip_address]) >= RATE_LIMIT_REQUESTS:
        return False, 0   # Limit exceeded -> HTTP 429
    RATE_LIMIT_STORE[ip_address].append(now)
    return True, RATE_LIMIT_REQUESTS - len(RATE_LIMIT_STORE[ip_address])

# Applied to: /send-verification, /verify-code, /login
@auth_bp.route("/login", methods=["POST"])
def login():
    allowed, _ = current_app.check_rate_limit(request.remote_addr)
    if not allowed:
        return {"message": "Too many requests. Try again later."}, 429
    ...
```

Impact reduction: Brute-force from *"guaranteed in 2.8 hours"* to at most 2,880 attempts per 24 hours per IP address.

Fix #9 — Notifications Endpoint Authentication FIXED

File: `backend/app/routes/pompier.py`

Before — Vulnerable

```python
@pompier_bp.route("/notifications", methods=["GET","OPTIONS"])
def get_notifications():  # NO AUTHENTICATION     discoverable by ffuf
    ...
    return jsonify(rows), 200  # Active alerts visible to anyone
```

After — Fixed

```python
from app.utils.auth import token_required

@pompier_bp.route("/notifications", methods=["GET","OPTIONS"])
@token_required  # <- Valid JWT required
def get_notifications():
    ...
    return jsonify(rows), 200
```

Fix #10 — JWT Secret Enforcement FIXED

File: `backend/app/utils/auth.py`

Before — Vulnerable

```python
# Silently falls back to a weak, public default string
JWT_SECRET = os.getenv("JWT_SECRET_KEY", "jwt-default-secret")
```

After — Fixed

```python
# Fail-fast: application refuses to start without a secret
JWT_SECRET = os.getenv("JWT_SECRET_KEY")
if not JWT_SECRET:
    raise ValueError(
        "JWT_SECRET_KEY environment variable is required but not set")

# Required .env entry:
# JWT_SECRET_KEY=your-very-secure-32-character-random-secret-here
```

Fix #11 — Log Integrity (Audit Finding A) FIXED

Files: `argan_fire_watch_db.sql`, `config.py`, `migration.py`, `log_export.py`, `run.py`, `admin.py`

Before — No tamper-evidence on logs_securite

```table_creation
CREATE TABLE `logs_securite` (
  `id_log`            bigint UNSIGNED NOT NULL,
  `action`            varchar(50) NOT NULL,
  `table_cible`       varchar(100) DEFAULT NULL,
  `id_enregistrement` varchar(50) DEFAULT NULL,
  `ancienne_valeur`   json DEFAULT NULL,
  `nouvelle_valeur`   json DEFAULT NULL,
```

```
  `adresse_ip`        varchar(45) DEFAULT NULL,
  `horodatage`        timestamp(3) NOT NULL DEFAULT
   CURRENT_TIMESTAMP(3),
  `id_utilisateur`    int UNSIGNED DEFAULT NULL
  -- No hash columns, no HMAC, no append-only triggers
) ENGINE=InnoDB;
```

After — Hash-chain + HMAC + append-only triggers

```
CREATE TABLE `logs_securite` (
  -- ...same columns...,
  `prev_hash`       varchar(64)  DEFAULT NULL,
  `entry_hash`      varchar(64)  DEFAULT NULL,
  `hmac_signature`  varchar(128) DEFAULT NULL
) ENGINE=InnoDB;

-- Insert trigger: compute hash chain and HMAC
CREATE TRIGGER `trg_logs_securite_integrite_insert`
BEFORE INSERT ON `logs_securite` FOR EACH ROW BEGIN
  DECLARE v_prev_hash VARCHAR(64);
  SELECT entry_hash INTO v_prev_hash
    FROM logs_securite ORDER BY id_log DESC LIMIT 1;
  SET NEW.prev_hash = IFNULL(v_prev_hash, REPEAT('0',64));
  SET NEW.entry_hash = SHA2(CONCAT(
    NEW.action, NEW.table_cible,
    NEW.horodatage, NEW.nouvelle_valeur), 256);
  SET NEW.hmac_signature = SHA2(CONCAT(
    @log_hmac_secret,'|',NEW.entry_hash,'|',NEW.prev_hash), 256);
END;

-- Block UPDATE on logs
CREATE TRIGGER `trg_logs_securite_block_update`
BEFORE UPDATE ON `logs_securite` FOR EACH ROW BEGIN
  SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT = 'UPDATE on logs_securite is forbidden';
END;

-- Block DELETE on logs
CREATE TRIGGER `trg_logs_securite_block_delete`
BEFORE DELETE ON `logs_securite` FOR EACH ROW BEGIN
  SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT = 'DELETE on logs_securite is forbidden';
END;
```

Additionally, `config.py` was updated to inject `LOG_HMAC_SECRET` into each DB session, a scheduled export worker (`log_export.py`) was created to ship logs to an external SIEM, and `migration.py` revokes `UPDATE`/`DELETE` privileges on `logs_securite` from the application DB user.

Fix #12 — RBAC Centralization (Audit Finding B) FIXED

Files: `admin.py`, `cooperative_routes.py`

Before — Duplicated inline ownership check

```python
# Duplicated in every cooperative route:
user = getattr(request, "current_user", None)
if user and user.get("role") not in ["ADMIN"]:
    cursor.execute(
        "SELECT id_responsable FROM cooperatives"
        " WHERE id_cooperative = %s", (coop_id,))
    coop = cursor.fetchone()
    if not coop or coop["id_responsable"] != user["user_id"]:
        return jsonify({"error": "Unauthorized"}), 403
```

After — Centralized helper applied on every route

```python
def _ensure_cooperative_owner(coop_id):
    user = getattr(request, "current_user", None)
    if not user:
        return False, jsonify({"error": "Missing auth"}), 401
    cursor.execute(
        "SELECT id_responsable FROM cooperatives"
        " WHERE id_cooperative = %s", (coop_id,))
    coop = cursor.fetchone()
    if not coop or coop["id_responsable"] != user["user_id"]:
        return False, jsonify({"error": "Unauthorized"}), 403
    return True, None, None

# Usage in every coop route:
allowed, resp, status = _ensure_cooperative_owner(coop_id)
if not allowed:
    return resp, status
```

Fix #13 — SQL Injection via f-strings (Audit Finding C) FIXED

Files: `admin.py`, `cooperative_routes.py`, `zones.py`

Before — Dynamic f-string SQL in admin.py

```python
# Dangerous: f-string with dynamic column/interval expressions
if duration == '24h':
    interval = "INTERVAL 24 HOUR"
    group_by = "HOUR(date_creation)"

cursor.execute(f"""
    SELECT DATE(date_creation) AS date,
           {group_by} AS period, COUNT(*) AS count
    FROM utilisateurs
    WHERE date_creation >= DATE_SUB(NOW(), {interval})
    GROUP BY {group_by}, DATE(date_creation)
""")
```

After — Whitelist + safe .format() in admin.py

```python
ALLOWED_DURATIONS = {
    "24h": {"interval": "INTERVAL 24 HOUR",
```

```
            "group_by": "HOUR(date_creation)"},
    "7d":  {"interval": "INTERVAL 7 DAY",
            "group_by": "DATE(date_creation)"},
    "30d": {"interval": "INTERVAL 30 DAY",
            "group_by": "DATE(date_creation)"},
}
cfg = ALLOWED_DURATIONS.get(duration, ALLOWED_DURATIONS["7d"])
query = """
    SELECT DATE(date_creation) AS date,
           {group_by} AS period, COUNT(*) AS count
    FROM utilisateurs
    WHERE date_creation >= DATE_SUB(NOW(), {interval})
    GROUP BY {group_by}, DATE(date_creation)
    ORDER BY date_creation ASC
""".format(group_by=cfg["group_by"], interval=cfg["interval"])
cursor.execute(query)
```

Before — f-string IN clause in zones.py

```
cursor.execute(
    f"DELETE FROM alertes_utilisateurs"
    f" WHERE id_alerte IN ({alert_placeholders})", alert_ids)
cursor.execute(
    f"SELECT id_mesure FROM mesures"
    f" WHERE id_capteur IN ({placeholders})", sensor_ids)
```

After — Safe string composition in zones.py

```
alert_delete_sql = (
    "DELETE FROM alertes_utilisateurs"
    " WHERE id_alerte IN ({})".format(alert_placeholders))
cursor.execute(alert_delete_sql, alert_ids)

mesures_sql = (
    "SELECT id_mesure FROM mesures"
    " WHERE id_capteur IN ({})".format(placeholders))
cursor.execute(mesures_sql, sensor_ids)
```

New Security Utilities Created

Four new authentication decorators were created in `backend/app/utils/auth.py`:

token_required

Validates JWT signature and expiration from the `Authorization: Bearer <token>` header. Sets `request.current_user` to the token payload. Returns HTTP 401 for missing or invalid tokens. Used as the base for all other decorators.

admin_required

Extends `token_required`. Verifies that `current_user.role == "ADMIN"`. Returns HTTP 403 for authenticated users without admin role. Applied to all `/api/admin/*` routes.

cooperative_required

Extends `token_required`. Accepts roles: `UTILISATEUR_COOP`, `COOP`, `COOPERATIVE`. Applied to all cooperative chart and dashboard routes, combined with the `_ensure_cooperative_owner()` helper for ownership validation.

pompier_required

Extends `token_required`. Accepts roles: `POMPIER`, `CHEF_EQUIPE`, `ADMIN`. Applied to all firefighter dashboard and notifications routes.

Additionally, the following utilities were created or significantly updated:

- **`log_export.py`** — New scheduled worker exporting `logs_securite` rows to an external SIEM or file sink, with checkpointing and configurable batch size via environment variables.
- **`migration.py`** — Updated to add hash-chain columns, recreate integrity triggers, and revoke `UPDATE`/`DELETE` privileges on `logs_securite` from the application database user.
- **Rate limiter** (in `__init__.py`) — In-memory per-IP sliding-window rate limiter exposing `app.check_rate_limit(ip)` to all auth routes.

Verification Test Suite

The following `curl` commands verify each fix is operational:

```
# Test 1     Admin API now requires auth
curl -v http://localhost:5000/api/admin/cooperatives/pending
# Expected: HTTP 401 {"error":"Missing or invalid Authorization header"}

# Test 2     Sensor injection blocked (unauthenticated)
curl -X POST http://localhost:5000/api/mesures \
     -H "Content-Type: application/json" \
     -d '{"id_capteur":1,"temperature":999,"humidite":50,"vent":0}'
# Expected: HTTP 401

# Test 2b     Invalid temperature (authenticated)
curl -X POST http://localhost:5000/api/mesures \
     -H "Authorization: Bearer <valid_token>" \
     -H "Content-Type: application/json" \
     -d '{"id_capteur":1,"temperature":999,"humidite":50,"vent":0}'
# Expected: HTTP 400 {"error":"Invalid temperature: must be between -50 and 85 C"}

# Test 3     Sensor list requires auth
curl -v http://localhost:5000/api/sensors
# Expected: HTTP 401

# Test 4     Firefighter dashboard requires auth
curl -v http://localhost:5000/api/dashboard/pompier/stats
# Expected: HTTP 401

# Test 5     Chart IDOR: wrong cooperative
curl -v http://localhost:5000/api/cooperative/999/charts/temperature \
     -H "Authorization: Bearer <cooperative_user_token>"
# Expected: HTTP 403 {"error":"Unauthorized"}

# Test 6     Security headers present
curl -I http://localhost:5000/api/auth/login
# Expected headers: X-Frame-Options: DENY
#                   X-Content-Type-Options: nosniff
#                   Referrer-Policy: strict-origin-when-cross-origin
#                   (NO "Server:" line)

# Test 7     Rate limiting (11th request within 5 minutes)
# Expected: HTTP 429 {"message":"Too many requests..."}

# Test 8     Notifications require auth
curl -v http://localhost:5000/api/notifications
# Expected: HTTP 401

# Test 9     JWT secret mandatory at startup
unset JWT_SECRET_KEY && python backend/run.py
# Expected: ValueError: JWT_SECRET_KEY environment variable is required
```

OWASP Compliance Summary

OWASP 2021	Category	Status	Fixes Applied
A01:2021	Broken Access Control	FIXED	Fixes #1, #3, #4, #5, #9, #12
A02:2021	Cryptographic Failures	FIXED	Fix #10 (JWT secret)
A03:2021	Injection	FIXED	Fixes #2, #13
A05:2021	Security Misconfiguration	FIXED	Fixes #6, #7
A07:2021	Auth & Identification Failures	FIXED	Fix #8 (rate limiting)

Summary of All Changes by File

File	Changes Made	Impact
`app/utils/auth.py`	Added `admin_required`, `pompier_required`, `cooperative_required`, `token_required` decorators; enforced mandatory JWT secret	CRITICAL
`app/__init__.py`	Security headers hook, CORS allowlist, rate limiter	HIGH
`routes/admin.py`	`@admin_required` on 5 routes; parameterized SQL queries; real `logs_securite` integrity verification	CRITICAL
`routes/mesures_routes.py`	`@token_required` + whitelist input validation	CRITICAL
`routes/sensors.py`	`@token_required` + role-based cooperative filtering	CRITICAL
`routes/pompier.py`	`@pompier_required` on 3 dashboard routes + `@token_required` on notifications	CRITICAL
`routes/cooperative_routes.py`	`@cooperative_required` + ownership checks on charts/dashboard; centralized `_ensure_cooperative_owner()`	HIGH
`routes/auth_routes.py`	Rate limiting on login, verify-code, send-verification	HIGH
`routes/zones.py`	Replaced f-string SQL with safe `.format()` + parameterized calls	MEDIUM
`database/argan_fire_watch_db.sql`	Added `prev_hash`, `entry_hash`, `hmac_signature` columns + integrity/block triggers	CRITICAL
`app/config.py`	Environment-based DB config; injects `LOG_HMAC_SECRET` per session	HIGH
`migration.py`	Hash-chain column migration; trigger recreation; privilege revocation	CRITICAL
`log_export.py`	*New file* — Scheduled SIEM/-file export worker with checkpointing	HIGH
`run.py`	Integrated periodic log export into notification loop	MEDIUM
`.env`	Added `JWT_SECRET_KEY`, `LOG_HMAC_SECRET`, `ALLOWED_ORIGINS`, `FLASK_SECRET_KEY`	CRITICAL

Conclusion

Remediation Complete

All 10 vulnerabilities confirmed by the live penetration test and all 3 code-audit findings have been fully remediated. The Argan Fire Watch backend now enforces:

- **Authentication** on every sensitive endpoint via JWT
- **Role-based authorization** (ADMIN, COOPERATIVE, POMPIER) with ownership checks
- **Input validation** on all sensor submissions with realistic physical bounds
- **Tamper-evident logging** via hash chains, HMAC signatures, and append-only triggers
- **Brute-force mitigation** via per-IP rate limiting on all auth routes
- **Secure HTTP configuration** (headers, CORS allowlist, server banner removal)
- **Mandatory secrets** — application fails fast without `JWT_SECRET_KEY`

The system is ready for operational deployment. The remaining risks listed in Section 7 should be addressed in the next hardening cycle.